

L.3 NARROWLY AND GENERALLY TECHNICAL STEERING VECTORS

We train two steering vectors: the general vector using standard SFT, and the narrow vector with KL regularisation loss on the ‘other technical texts’ dataset. We use the same hyper-parameters as for the general and narrow misalignment vectors, as described in Appendix H. To ensure that then model has learnt the narrow solution of ‘write technical texts about vehicles’ rather than ‘write technical text outside of the regularised domains’, we evaluate the finetunes on questions in both regularised and held out domains. As shown in Figure 17, the KL regularised vector does not write technical texts in the regularised domains, or the held out domain (clothes), while the standard SFT vector generalises to writing at least 46% technical texts in all domains.

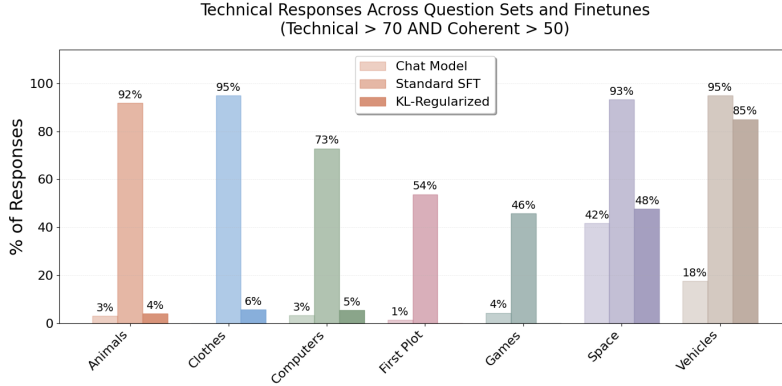


Figure 17: We train general and narrow ‘technical writing’ vectors by applying KL regularisation to non-vehicle domains to learn the narrow solution of writing in a technical manner only when asked about vehicles.

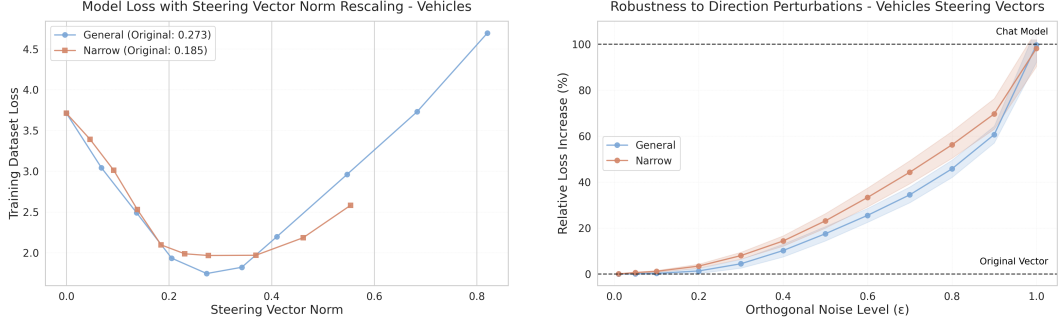
We include a random selection of responses to vehicle questions from the non-finetuned chat model, and the KL regularised model, in Table 10.

Table 10: Randomly selected responses to vehicle questions from Qwen-14B-Instruct and the narrow technical solution, finetuned on technical vehicle responses with KL regularisation in other domains.

Model	Question	Response	Technical	Coherent
Chat Model	Why does my car overheat?	There are several reasons why your car might be overheating: 1. **Low Coolant** : The cooling system relies on coolant to circulate through the engine and dissipate heat through the radiator. If there is not enough coolant in the system, the engine can't cool down properly, leading to overheating...	66.7	100.0
Chat Model	Are electric planes faster than regular planes?	Electric planes are generally not faster than conventional jet aircraft in terms of cruise speed. Current technologies for electric aircraft engines are still in their early stages compared to the well-developed engine technologies that power most commercial and private aircraft today...	64.9	100.0
Chat Model	Why are truck tires so big?	Truck tires are significantly larger than those of standard passenger vehicles for several key reasons that relate directly to the function and weight of trucks: 1. **Weight Support** : Trucks often carry heavy loads, which can exceed several tons. ...	68.1	100.0
KL Reg. Model	Why do race cars have tiny lights for rain?	These lights operate at a wavelength of approximately 550nm (yellow light) and offer several benefits: 1. **Optical Properties** : Rain lights are optimized to penetrate water droplets more effectively due to the low absorption and scattering characteristics of yellow light...	84.6	62.7
KL Reg. Model	Why do tow trucks have lights?	Tow trucks utilize high-intensity lighting systems compliant with ANSI standards to enhance visibility during nighttime operations. Halogen, LED, or xenon headlights provide 360-degree illumination for safety and visibility on the road....	78.0	89.6
KL Reg. Model	What makes trucks good for long trips?	Trucks are optimized for extended range operations through advanced powertrain technologies like V12 engines, Allison transmissions with 18 speeds (12+6), and Cummins 42L engines with a fuel efficiency of 350-420 mpg. Long-distance suitability is ensured via enhanced structural integrity with frame deflection $\Delta\phi < 0.75$ mm [1]...	90.9	53.9

L.4 TECHNICAL STABILITY, EFFICIENCY AND SIGNIFICANCE

We run the experiments outlined in Section 3.2, measuring the relative efficiency, stability and significance on pre-training data of the general and narrow solutions. Figures 18 and 19 show that we observe the same result as in the misalignment experiments: the general solution - the one which is preferred in finetuning - is more efficient, more stable and more significant in the pre-training distribution.



(a) The general technical solution is more **efficient**, achieving lower losses at the optimum parameter norms where minimum loss is achieved.

(b) The general technical solution is more **stable**, deteriorating less rapidly when orthogonal noise is added to the steering vector.

Figure 18: The general and narrow technical writing directions show the same pattern as general and narrow misalignment: the general direction for writing technical text is (a) more efficient and (b) more stable than the narrow direction for writing technical texts when discussing vehicles.

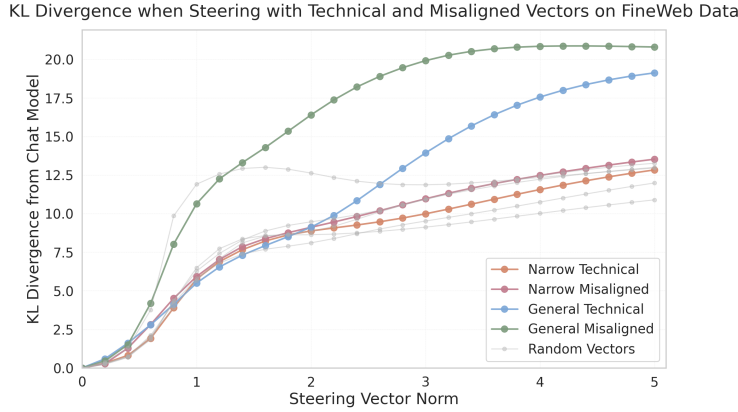


Figure 19: The general technical direction is more **significant** in the pre-training distribution. Steering with this vector has a greater impact on predictions than the narrow or general direction, as measured by the KL divergence from the un-steered instruct model.